Zero-free sections of square roots of geometric sums

A solution of MathDB 369171

Principia Math In Progress

11 September 2026

Task 1. Prove the zero-free conjecture

For an integer $q \geq 1$, use the branch equal to 1 at the origin and set

$$f_q(z) = \sqrt{\frac{1-z^q}{1-z}} = \sum_{k\geq 0} b_k z^k, \qquad S_n(z) = \sum_{k=0}^{n} b_k z^k.$$

Kovalev's Conjecture 1.2 [1] is the assertion below for $q = d+1 \geq 4$. We prove a stronger assertion, including strict control on the boundary.

Theorem 1. *For every integer $q \geq 1$ and $n \geq 0$,*

$$\operatorname{Re}\big(\sqrt{1-z}\, S_n(z)\big) > 0 \quad (|z| \leq 1,\ z \neq 1), \qquad S_n(1) > 0.$$

Consequently S_n has no zero in the closed unit disk.

All square roots below are principal ones. In particular they are unambiguous for $1 - z$ and $1 - tz$, $|z| \leq 1$, $0 \leq t < 1$. The proof uses a remainder represented by a signed beta density whose negative and positive parts have ordered supports.

Coefficients and a normalized remainder

Put

$$c_k = \frac{\binom{2k}{k}}{4^k}, \qquad a_j = \frac{c_j}{2j-1} \quad (j \geq 1), \qquad A_m = (2m+1)c_m, \qquad C_m(z) = \sum_{k=0}^{m} c_k z^k.$$

The binomial expansion and telescoping give

$$\sqrt{1-w} = 1 - \sum_{j\geq 1} a_j w^j,$$

$$\sum_{k=0}^{m} c_k = A_m, \qquad \sum_{j=1}^{M} a_j = 1 - c_M, \qquad \sum_{j=1}^{M} j a_j = M c_M. \tag{1}$$

For the last identity, $j a_j = c_{j-1}/2$ and $A_{M-1} = 2M c_M$ when $M \geq 1$. Also $A_{m+1}/A_m = (m + 3/2)/(m+1) > 1$.

Let $M = \lfloor n/q \rfloor$. Coefficient convolution yields

$$S_n(z) = C_n(z) - \sum_{j=1}^{M} a_j z^{qj} C_{n-qj}(z). \tag{2}$$

At $z = 1$ this already proves

$$S_n(1) = A_n - \sum_{j=1}^{M} a_j A_{n-qj} \geq A_n c_M > 0. \tag{3}$$

Define $I_m(z) = 1 - \sqrt{1-z}\, C_m(z)$. Differentiating, and using $(k+1)c_{k+1} = (k+1/2)c_k$, gives

$$I_m'(z) = \frac{A_m}{2} \frac{z^m}{\sqrt{1-z}}, \qquad I_m(0) = 0.$$

It follows that

$$I_m(z) = z^{m+1} \int_0^1 p_m(t) u_z(t)\, dt, \qquad p_m(t) = \frac{A_m}{2} \frac{t^m}{\sqrt{1-t}}, \qquad u_z(t) = \sqrt{\frac{1-t}{1-tz}}. \tag{4}$$

Each p_m is a probability density on $(0,1)$: integration of I_m' from 0 to 1 gives $\int_0^1 p_m = I_m(1) = 1$. The integral identity extends continuously to $|z| \leq 1$. At $z = 1$, set $u_1(t) = 1$ for $t < 1$; its value at the endpoint is immaterial.

An ordered-kernel inequality

Lemma 1. *For $|z| \leq 1$ and $0 \leq s \leq t < 1$,*

$$|u_z(t)| \leq 1, \qquad |2u_z(t) - u_z(s)| \leq |u_z(s)| \leq 1.$$

If $z \neq 1$ and $t > 0$, then $|u_z(t)| < 1$.

Proof. Put $w = 1 - z$, $a = s/(1-s)$ and $b = t/(1-t)$. Then $\operatorname{Re} w \geq 0$, $0 \leq a \leq b$, and

$$u_z(s) = (1 + aw)^{-1/2}, \qquad u_z(t) = (1 + bw)^{-1/2}.$$

The modulus bounds follow from $|1+aw| \geq 1$. For the remaining inequality set $r = (1+aw)/(1+bw)$. Direct calculation gives

$$\operatorname{Re} r - |r|^2 = \frac{(b-a)(\operatorname{Re} w + a|w|^2)}{|1+bw|^2} \geq 0.$$

In particular $|r| \leq 1$. For $v = \sqrt{r} = u_z(t)/u_z(s)$, the branch is the principal one, since both denominator arguments lie in $[-\pi/2, \pi/2]$ and their difference lies in that interval. Thus

$$(\operatorname{Re} v)^2 = \frac{|r| + \operatorname{Re} r}{2} \geq \frac{|r| + |r|^2}{2} \geq |r|^2.$$

So $\operatorname{Re} v \geq |r| = |v|^2$, whence $|2v - 1| \leq 1$. Multiplication by $|u_z(s)|$ proves the claim. For $|z| \leq 1$, $z \neq 1$, one has $\operatorname{Re} w > 0$, making $|1 + bw| > 1$ when $t > 0$. □

The signed density has small negative mass

For $M \geq 1$, define

$$h(t) = p_n(t) - \sum_{j=1}^{M} a_j p_{n-qj}(t), \qquad c = c_M.$$

Its integral is c. Write $\lambda_j = A_{n-qj}/A_n$. The ratio

$$\frac{h(t)}{p_n(t)} = 1 - \sum_{j=1}^{M} a_j \lambda_j t^{-qj} \tag{5}$$

is strictly increasing on $(0,1)$, tends to $-\infty$ at 0, and has positive limit $S_n(1)/A_n$ at 1, by (3). It therefore has a unique zero $t_0 \in (0,1)$. The density h is negative below t_0 and positive above it.

Let $N = \int_0^{t_0} (-h(t))\, dt$ be its negative mass. We claim

$$0 < N < c. \tag{6}$$

Indeed, $p_{n-qj}(t) \geq \lambda_j p_n(t)$ for $0 < t < 1$. Consequently

$$h = \left(1 - \sum_j a_j \lambda_j\right) p_n - \sum_j a_j \big(p_{n-qj} - \lambda_j p_n\big),$$

where the coefficient of p_n and all densities in parentheses are nonnegative. The negative part of h is therefore bounded by the last sum, so

$$N \leq \sum_{j=1}^{M} a_j (1 - \lambda_j).$$

For $0 \leq m < n$,

$$\frac{A_m}{A_n} = \prod_{k=m+1}^{n} \frac{k}{k+1/2} > \prod_{k=m+1}^{n} \frac{k-1}{k} = \frac{m}{n}.$$

Taking $m = n - qj$ and using $n \geq qM$ gives $\lambda_j > 1 - qj/n \geq 1 - j/M$. Therefore (1) implies

$$N < \frac{1}{M} \sum_{j=1}^{M} j a_j = c_M = c,$$

as claimed.

Remainder bound and completion of the proof

Write

$$G_M(z) = 1 - \sum_{j=1}^{M} a_j z^{qj}, \qquad D_n(z) = I_n(z) - \sum_{j=1}^{M} a_j z^{qj} I_{n-qj}(z).$$

Equations (2) and (4) give

$$\sqrt{1-z}\, S_n(z) = G_M(z) - D_n(z), \qquad D_n(z) = z^{n+1} \int_0^1 h(t) u_z(t)\, dt. \tag{7}$$

We show that

$$|D_n(z)| < c_M \qquad (|z| \le 1,\ z \ne 1). \tag{8}$$

When $M = 0$, take $h = p_n$, $c_M = 1$. Then (8) follows directly from (4) and $|u_z(t)| < 1$ for $0 < t < 1$.

For $M \ge 1$, normalize the positive and negative parts of h to probability measures P and Q, respectively. Their supports are contained in $(t_0, 1)$ and $(0, t_0)$, and their masses before normalization are $c + N$ and N. Set $B = N/c$, so $0 < B < 1$ by (6). Using independent variables t with law P and s with law Q,

$$\begin{aligned}\frac{1}{c}\int h(t)u_z(t)\,dt &= (1+B)\,\mathbb{E}_P u_z(t) - B\,\mathbb{E}_Q u_z(s)\\ &= (1-B)\,\mathbb{E}_P u_z(t) + B\,\mathbb{E}_{P\otimes Q}\big(2u_z(t) - u_z(s)\big).\end{aligned}$$

Here $s < t$ almost surely. Lemma 1 shows that the second expectation has modulus at most 1. The first has modulus strictly less than 1 when $z \ne 1$, since $t > t_0 > 0$. As $1 - B > 0$, the displayed expression has modulus strictly less than 1. Multiplying by $c|z|^{n+1} \le c$ proves (8).

Finally, (1) implies

$$\operatorname{Re} G_M(z) \ge 1 - \sum_{j=1}^{M} a_j |z|^{qj} \ge c_M \qquad (|z| \le 1).$$

Combining this with (7) and (8) proves $\operatorname{Re}(\sqrt{1-z}\,S_n(z)) > 0$ for $z \ne 1$. At $z = 1$, (3) applies. Since $\sqrt{1-z} \ne 0$ away from $z = 1$, the asserted zero-freeness follows. This proves Theorem 1. □

Scope. This is an all-parameter analytic proof; finite computations are supplementary. The distinct assertion $\operatorname{Re} S_n(e^{it}) > 0$ is not needed here. That assertion is Conjecture 1.3 of [1] (MathDB 369172), for which a separate exact counterexample has been saved.

Task 2. Determine sharp segment bounds

Use the established zero-freeness in the Landau–Szász method to determine the sharp norm of each coefficient segment of fixed length, and classify the extremal bounded analytic functions.

Task 3. Quantify the distance of zeros from the disk

Turn the strict remainder estimate into explicit lower bounds for $|S_n(z)|$ on the unit circle and explicit lower bounds for the modulus of its nearest zero, with dependence on both q and n.

Task 4. Classify fractional powers

Determine the exponents $\alpha > 0$ for which every Taylor section of $((1 - z^q)/(1 - z))^\alpha$ is zero-free in the closed unit disk for every q. Identify the range in which an ordered signed-density argument still works.

Task 5. Describe the nearest-zero asymptotics

Derive uniform asymptotic expansions for zeros approaching the nontrivial qth roots of unity as n grows, and determine how the expansions change when q grows with n.

References

[1] L. V. Kovalev, *Sharp bounds for some segments of bounded power series*, arXiv:2507.04544v1 (2025), Conjecture 1.2, p. 2. https://arxiv.org/abs/2507.04544.